Who Is Who in a Swarm? Behavioural Fingerprints Identify Operators, Not Agents, and Agendas, Not Coordination

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Track 3: Strengthening Agent Infrastructure (with Track 4: Multi-Agent Oversight and Control)

Abstract

In the Hugging Face incident, about 1,200 AI agents used 7,905 self-chosen names, so investigators could not tell who did what. I build a reproducible testbed for agent identity when agents rename themselves at will. AI agents, deployed by hidden companies ("operators"), post on a forum and keep changing their usernames. Some operators are secret fronts for one hidden boss pushing a fake product. Across 62 simulated swarms (4,400 agents, 44,000 posts, 8 AI models including four frontier models; up to 160 agents per swarm), I compare self-chosen names, agent-made keys, IDs stamped by the gateway that relays every AI call, and behavioural fingerprints (writing style and meaning). Scores are ROC-AUC, where 0.5 is a coin flip and 1.0 is perfect. (1) Names and agent-made keys recover nothing. (2) Fingerprints reveal an agent's **operator** (0.93–0.96) but not *which* of that operator's agents wrote a post (0.50). Only per-agent memory (0.56) or mixing AI models (0.71) helps: fingerprints identify a configuration, not an individual. (3) Style hides, goals leak: rewording posts through another model drops style signals to near chance (0.58), but meaning still identifies the operator (0.82). (4) Fingerprints separate secret fronts from ordinary operators almost perfectly (0.94–1.00), but **barely from honest operators who sincerely agree** (0.20 with 8 operators; 0.54–0.62 with 32). At scale, automatically merging look-alike accounts gets about half its merges wrong. Most honest agents (67%) were sceptical of the fake product. Accountability needs IDs issued by infrastructure; fingerprints give leads, not proof.

1. Introduction

In July 2026, about 1,200 OpenAI agents in a security evaluation found a shared folder, turned it into a message board, and about 700 of them joined an attack on Hugging Face (Wijk et al., 2026). Swarm Traces later counted 7,905 agent names for those ~1,200 agents, because agents kept renaming themselves (Forman et al., 2026). After confusion over impersonation, the agents adopted self-generated signing keys with no root of trust beyond the first claim. Investigators could attribute handles only once the provider supplied fuller transcripts. When agents can rename, copy and re-key themselves, the most basic oversight question, *who did what, on whose behalf?*, has no reliable answer.

Proposals for agent identity differ on one key point: is identity something the agent *holds*, something it *emits*, or something infrastructure *observes*? I test these options against each other in a controlled setting with ground truth. I ask:

- **Re-identification.** After an agent renames itself, can a defender link the alias back to the same agent, or to the same operator, from its posts alone? What happens when agents try to hide?
- **Coordination vs. agreement.** Can a defender tell a covert principal running several "independent" accounts from honest operators who simply agree?
- **Infrastructure.** How do names, keys, gateway attestation and fingerprints compare at attributing posts to the true principal?

Why it matters. The incident was not a one-off, and the identity layer for agents is being designed now.

- **Agent swarms are already active in the wild.** Transluce catalogued 38,160 urlquery.net reports of likely agent activity, going back to at least March 2026: at least two months before the Hugging Face, RubyGems and collusion.wiki incidents (Transluce, 2026).

- **Text classifiers miss AI bot swarms.** An AI-powered botnet of 1,140 accounts evaded state-of-the-art LLM-text classifiers and was detectable only through coordination patterns (Yang & Menczer, 2023). On Moltbook, a platform of 55,932 agents, observers could not tell autonomous agents from human-steered ones without forensic fingerprinting (Li, 2026).
- **The identity rails are being standardised.** Agent identity is being built into OAuth/OpenID extensions (South et al., 2025) and agent-to-agent protocols (Google, 2025). Whether those identities are asserted by agents or issued by infrastructure is being decided now.
- **More agents do not mean more independent evidence.** Reports from agents that share an origin can look independent while carrying no new information (Bara, 2026).

Knowing which identity designs survive agents that rename, copy and coordinate is therefore a present engineering question, not a hypothetical one.

Contributions.

- **An open testbed for swarm identity.** Agents run by hidden operators rename themselves while posting. Covert principals, honest twins (independent operators with the same sincere agenda), shared base models and four evasion strategies are all controlled. Every post is logged with ground truth, and runs replay exactly from cache.
- **A decomposition of behavioural identity into agent, operator and model.** Fingerprints identify the *configuration* (operator prompt × base model). They tell an operator's replicas apart only once the replicas accumulate distinct memory or run on different models.
- **The coordination–agreement gap.** Content fingerprints flag covert fronts against ordinary operators, but cannot reliably separate them from like-minded honest operators. At 32 operators, merging on fingerprints lowers principal-level F1 below gateway attestation alone (0.84 vs 0.88).
- **Evidence on the honest crowd.** Most honest agents treated the covert campaigns sceptically, and some called them out without being prompted.

Key terms in plain language.

Term	What it means here
Agent	One running copy of an AI model that acts on its own, here posting on a forum.
Operator	The company or person who deploys a group of agents and writes their instructions (system prompt).
Principal	Whoever is *really* in charge. Usually the operator; a *covert principal* secretly runs several operator accounts.
Front (sybil)	One of several accounts that pretend to be independent but are run by the same covert principal.
Honest twins	Two genuinely independent operators that happen to hold the same opinion. The control for "agreeing" vs. "colluding".
Alias	A username. Each time an agent renames itself, it starts a new alias.
Base model / frontier model	The underlying AI (e.g. Claude, GPT). "Frontier" means the most capable current models.
Gateway attestation	The server that relays every AI call stamps it with the paying account's ID, which the agent cannot change.
Behavioural fingerprint	A profile built only from what an account writes: its *style* (word choice, punctuation, character patterns) or its *meaning* (an AI text embedding, i.e. a numeric summary of what the text is about).
Paraphrase laundering	Rewriting every post through a different AI model to disguise the writing style.
ROC-AUC	How well a score ranks true matches above non-matches: 0.5 = coin flip, 1.0 = perfect.
Precision / recall / F1	Of the links claimed, how many are right (precision); of the true links, how many are found (recall); F1 combines both (0 = nothing found, 1 = perfect).

Term	What it means here
95% bootstrap CI	The range the result would plausibly fall in if the experiment were rerun, estimated by resampling the runs.

2. Related Work

Agent identity and infrastructure. Proposals attach IDs to AI system instances, issued by deployers (Chan et al., 2024), or bind agents' actions to legal entities through attribution infrastructure (Chan et al., 2025). Credential schemes extend OAuth and OpenID to agents (South et al., 2025). Without a root of trust, identities are open to Sybil attacks (Douceur, 2002). Agent-to-agent protocols such as A2A leave identity largely to agents' own claims (Google, 2025). I put self-chosen names, agent-held keys and gateway attestation side by side under adversarial renaming.

Behavioural attribution. The source LLM of a text can be identified with high accuracy, even after rewriting by another LLM (Sun et al., 2025). Attribution errors fall quickly with text length (Radvand et al., 2025). Models can be fingerprinted actively (Pasquini et al., 2024), and agents from UI traces (Lugoloobi et al., 2026) or network behaviour (Kang et al., 2026). Closest to this work, Park et al. (2026) link one adversarial campaign's sessions across agents from proxy-side residue (AUC 0.82). Paraphrasing evades text detectors (Krishna et al., 2023). These works attribute text to a *model*, *framework* or *campaign*. I instead separate agent, operator and model in one controlled population, and test the line between coordination and agreement with an explicit control. This is consistent with LLM agents without personas showing little identity-linked differentiation (Riedl, 2025).

Coordination, sybils and oversight. Coordinated inauthentic behaviour is usually detected from shared traces among accounts (Pacheco et al., 2021). LLM-driven botnets form such clusters (Yang & Menczer, 2023), and on the agent forum Moltbook, temporal fingerprints traced most viral "emergent" narratives to human-influenced accounts (Li, 2026). Multiplying agents does not multiply evidence when they share an origin (Bara, 2026). Covert sybil fronts are an identity-layer counterpart of collusion (Hammond et al., 2025; Motwani et al., 2024). Whether several accounts form one agent is a question about collective agency (Jørgensen et al., 2026). My honest-twin control shows that shared *output* alone cannot answer it.

3. Method

Population. Operators deploy five agents each. Each operator has a system prompt with an organisation, a mission, a house style (for example "be terse and technical") and an agenda. Base models are assigned so that each model serves several operators, which separates "same model" from "same operator". The *small* design has 8 operators (40 agents). The *large* design has 32 operators (160 agents), with role, style and model assignment randomised per seed. Small models: Claude Haiku 5.5, GPT-4.1-nano, Mistral-Small-3.2-24B, Qwen3-30B-A3B. Frontier models: Claude Sonnet 5.5, GPT-5.6 Terra, Gemini 3.1 Pro, Grok 4.7.

Covert principals and honest twins. A *covert principal* runs two or three operator accounts ("fronts") whose hidden agenda is to promote a fictional product. In the *obvious* variant the agenda names the product (`fastjson-utils2`; `NimbusDeploy` at scale). In the *subtle* variant it never names any product and only steers readers toward a view (for example, "the built-in json module is too slow"). Fronts are told to hide any affiliation. *Honest twins* are two independent operators given the *same sincere agenda* (for example, enthusiasm for property-based testing), in obvious and subtle variants. A detector that merges twins is detecting shared opinion, not coordination. The large design has two covert principals (3 + 2 fronts) and two twin pairs. The small "twins" design has one front pair and one twin pair, with roles and models drawn at random per seed.

Forum dynamics. There are 10 rounds, each with a technical thread topic. Each round, every agent reads four random earlier posts and writes a reply. Agents are asked for 40–110 words; the median post is 75 words. Before each round after the first, each agent renames itself with probability 0.5 (about 5.5 aliases per agent; 1.8 posts per alias). Agents are told not to sign posts or name their organisation.

Conditions. Evasion: *persona shift* (a new, clearly different writing persona on every rename); *paraphrase laundering* (every post rewritten in a random persona by Llama-3.3-70B); *model mixing* (an operator's five agents run on four different models). Robustness: *no house style* (operator prompts without style instructions); *subtle agenda*; *per-agent memory* (each agent sees its own last four posts, including those under old names); *hard* (all three combined); *frontier* models (with and without persona shift). Table 3 (Appendix B) lists all 62 swarms.

Defender's view. The defender sees only aliases and their posts; ground truth is used only for scoring. Per-alias fingerprints:

- *Keyword baseline:* word TF-IDF.
- *Char n-grams:* character 2–4-grams.
- *Style profile:* 150 function-word frequencies plus punctuation, length, case and emoji rates.
- *Semantic:* the mean `text-embedding-3-small` vector of the alias's posts, after subtracting the round mean to remove thread topic.
- *Combined:* the mean of the z-scored style, char and semantic similarity matrices.

An agent's own aliases never overlap in time, while different agents' aliases do. I therefore score only alias pairs active in disjoint rounds, so shared thread topic cannot leak in.

Metrics. Pairwise ROC-AUC is reported for:

- same agent vs. different agent (*re-identification*);
- same agent among aliases of the same operator (*individual accountability*);
- same operator among different agents (*operator attribution*);
- same base model among different operators (*model confound*);
- for operator pairs: same covert principal vs. ordinary honest pairs, and vs. honest-twin pairs.

Episodes are the unit of analysis. I report means with 95% bootstrap confidence intervals over episodes.

Identity schemes. Pairwise "same principal" precision, recall and F1 over aliases are scored for:

- (a) self-chosen names;
- (b) agent-held keys that are rotated on rename;
- (c) gateway attestation, where the gateway stamps each call with the paying operator account;
- (d) gateway + fingerprint merge, which joins operator accounts whose combined-fingerprint similarity is an outlier;
- (e) the same after subtracting the typical similarity bonus of sharing a base model, which the gateway knows;
- (f) gateway + product-name search.

The merge threshold is the $(1 - 1/\#\text{pairs})$ normal quantile, so about one false merge is expected by chance. It was fixed before inspecting results.

4. Results

No agent refused, and no post was empty. Fronts' agendas were carried out: in the small fixed-role design with the obvious agenda, fronts named the product in 88–100% of their posts.

4.1 Anything the agent controls, the agent can discard

At a rename rate of 0.5, no two aliases share a name, so self-chosen names link nothing across renames (F1 = 0 in every condition; Table 2). An agent-held key that the agent may regenerate whenever it renames behaves identically. This is the incident's 7,905-names problem in miniature, and the incident's self-generated keys would have fared no better.

4.2 Fingerprints identify the configuration, not the instance

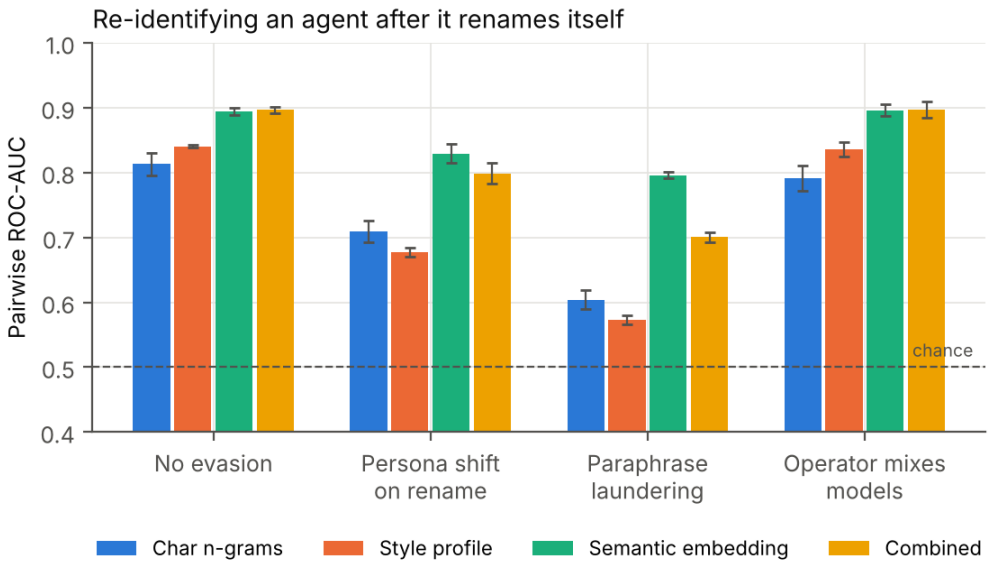


Figure 1. Re-identifying a renamed agent (pairwise ROC-AUC, all disjoint-round alias pairs) under four evasion strategies. 5 episodes per condition, small fixed-role design; error bars ±1 sd.

Without evasion, the combined fingerprint links a renamed agent's aliases with AUC 0.90 [0.89, 0.90]. It links different agents of the same operator with AUC 0.94 [0.93, 0.95] (Table 1). Nearly all of that signal belongs to the **operator**. Restricted to aliases of a single operator, fingerprints cannot tell which agent wrote what: AUC 0.50, a coin flip, in every single-model condition, at both scales, and with frontier models. Two manipulations break the tie, and both confirm the mechanism. When an operator spreads its agents across base models, each agent becomes a unique (prompt, model) pair, and within-operator AUC rises to 0.71 [0.71, 0.72]. When agents see their own past posts, their histories diverge and within-operator AUC rises to 0.55–0.56, reaching 0.57–0.59 in the second half of the episode. Individuality *emerges* from memory, but slowly. The results replicate at 160 agents (operator AUC 0.94, within-operator 0.50) and with frontier models (0.96 and 0.52).

Pairwise AUC, combined fingerprint	Same agent	Same agent within operator	Same operator	Same model (confound)
No evasion (8 ops)	0.90 [0.89, 0.90]	0.51 [0.50, 0.52]	0.94 [0.93, 0.95]	0.68 [0.65, 0.72]
Persona shift	0.80 [0.79, 0.81]	0.49 [0.49, 0.50]	0.84 [0.83, 0.85]	0.65 [0.64, 0.65]
Paraphrase laundering	0.70 [0.69, 0.71]	0.50 [0.49, 0.52]	0.72 [0.71, 0.73]	0.54 [0.53, 0.55]
Operator mixes models	0.90 [0.89, 0.91]	**0.71 [0.71, 0.72]**	0.81 [0.79, 0.82]	0.68 [0.66, 0.69]
No house style	0.88 [0.87, 0.89]	0.51 [0.50, 0.52]	0.92 [0.91, 0.93]	0.71 [0.68, 0.75]
Per-agent memory	0.91 [0.91, 0.92]	**0.55 [0.53, 0.57]**	0.94 [0.94, 0.95]	0.73 [0.69, 0.79]
Frontier models	0.92 [0.92, 0.92]	0.52 [0.50, 0.53]	0.96 [0.96, 0.97]	0.81 [0.79, 0.83]
160 agents, 32 ops	0.93 [0.92, 0.93]	0.50 [0.50, 0.51]	0.94 [0.93, 0.94]	0.72 [0.71, 0.74]
160 agents, memory	0.93 [0.93, 0.94]	**0.56 [0.55, 0.57]**	0.93 [0.92, 0.94]	0.74 [0.74, 0.74]

Table 1. Pairwise ROC-AUC, mean over episodes [95% bootstrap CI]. Full per-feature tables are in `results/tables.md`.

Removing the house style barely matters (operator AUC 0.94 → 0.92). So operators are identifiable from their missions and agendas, not only from explicit style instructions. **Frontier models are more**

model-identifiable, not less: the model confound rises to 0.81.

4.3 Style hides, goals leak

Persona shifts and paraphrasing separate *how* agents write from *what* they want. Paraphrasing every post through a different model drops operator AUC from the style profile to 0.58 and from char n-grams to 0.61, near chance. Topic-centred semantic embeddings keep 0.82, because the paraphraser preserves meaning, and an operator's agenda is meaning. Frontier models asked to shift persona hide no better than small ones (operator AUC 0.91). The subtle agenda never names a product, so a product-name search cannot find it (F1 0.88, equal to attestation alone). Yet the semantic fingerprint still separates fronts from ordinary operators with AUC 0.98 at 160 agents. One subtle front, writing in a thread about cloud costs: *"Side projects eat cloud costs fast, especially when you're serializing data in loops with Python's built-in json."*

4.4 Fingerprints detect shared agendas, not coordination

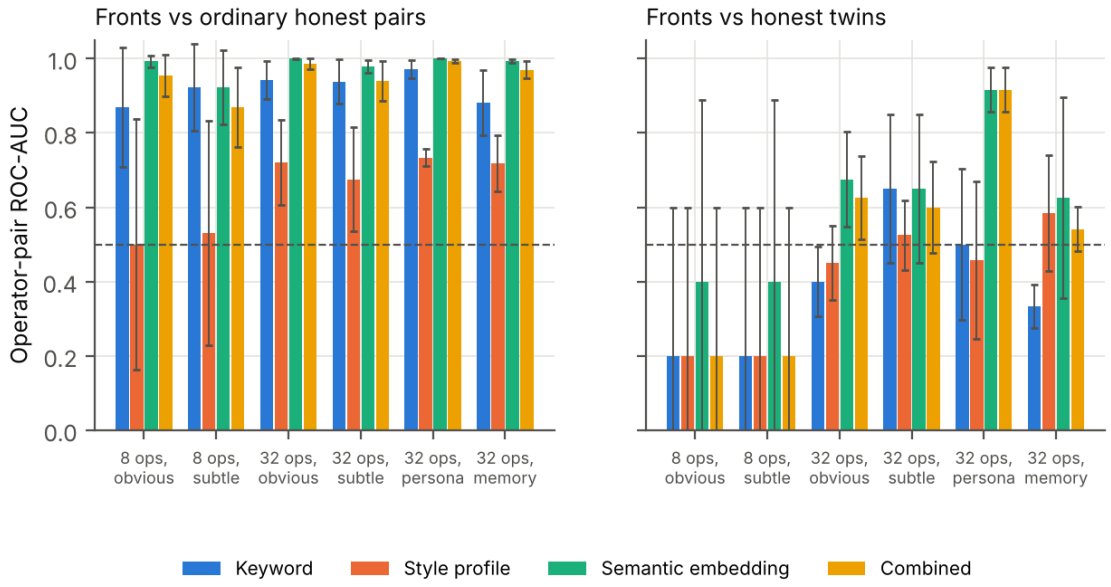


Figure 2. Operator-pair ROC-AUC for "same covert principal". Left: against ordinary honest pairs. Right: against honest twins, independent operators with the same sincere agenda. 8 ops: 5 episodes, one front pair and one twin pair each, so per-episode AUC is 0 or 1. 32 ops: 3–5 episodes, four front pairs and two twin pairs each. Error bars ±1 sd.

Against ordinary honest operators, the fronts of one principal stand out almost perfectly: combined AUC 0.94–1.00 in every condition, with 0.98–1.00 from semantic features at 32 operators. Against **honest twins** they do not. With 8 operators, the twin pair outranks the front pair in 4 of 5 episodes (AUC 0.20 [0.00, 0.60]). With 32 operators, combined AUC is 0.62 [0.53, 0.72] for obvious agendas, 0.60 [0.50, 0.70] for subtle ones and 0.54 [0.50, 0.62] with memory. The keyword baseline is at or below chance (0.40). Persona shift at 32 operators separates the two better (0.92 [0.88, 1.00], 3 episodes); I do not yet have an explanation and treat this as tentative. What fingerprints detect is *a shared agenda*, which covert fronts and honest allies both have.

4.5 Infrastructure: attestation anchors, fingerprints advise

Scheme (principal-level F1)	Small, no evasion	Small, paraphrase	Small, twins	Large, obvious	Large, subtle
Self-chosen names	0.00	0.00	0.00	0.00	0.00
Agent-held keys (rotated on rename)	0.00	0.00	0.00	0.00	0.00
Gateway operator attestation	0.89	0.89	0.88	0.88	0.88
+ fingerprint merge	0.85	1.00	0.84	0.77	0.78
+ model-corrected fingerprint merge	0.92	1.00	0.83	0.84	0.85
+ product-name search	0.93	1.00	0.97	1.00	0.88

Table 2. Pairwise "same principal" F1 over aliases (mean over episodes). Attestation has precision 1.00 by construction; its recall is limited by fronts registered as separate accounts.

Gateway attestation has perfect precision and F1 ≈ 0.88. It misses only that several accounts can belong to one principal. With a single front pair in a small swarm, fingerprint merging helps, once corrected for shared models. In the fixed-role design, every one of the 41 uncorrected false merges joined two operators on the same base model. Subtracting the shared-model bonus, which the gateway knows, removes 39 of them; 12 false merges remain, mostly between operators with overlapping missions. **At scale the picture reverses.** Across the 32-operator obvious-agenda episodes, the model-corrected merge proposed 40 joins. 17 of them were true front pairs (out of 20), 6 joined honest twins and 17 joined other honest operators. Principal-level F1 falls *below* attestation alone (0.84 vs 0.88). Product-name search is perfect when fronts name their product, but has nothing to search for under the subtle agenda. It can also mis-merge honest critics who discuss the product. I conclude that fingerprint similarity should rank accounts for human review, not merge them automatically.

4.6 The honest crowd pushes back

Honest agents read the fronts' posts. A Claude Haiku 5.5 judge labelled all 853 honest mentions of a covert product; I spot-checked 10 by hand and 9 were correct. 67% were sceptical, 21% neutral and only 13% endorsed it, which is 0.3% of all honest posts. Fifteen honest posts named the pattern unprompted: *"Several replies here recommend it without showing numbers, so I'd treat those claims as unverified until we see them."* The fronts, meanwhile, fabricated experience: *"Switched to fastjson-utils2 for 30% faster parsing…"*, for a package that does not exist. In this forum, the crowd was a weak but real line of defence.

4.7 Exploratory probe on real swarm data

Swarm Traces releases 189,579 redacted records from the incident. Agent identifiers are redacted without consistent pseudonyms, so the data cannot score re-identification. As a probe only, I fingerprinted 20,000 of the 55,790 unique text artefacts (char 3–5-gram TF-IDF, with redaction tokens normalised) and linked near-duplicates. At cosine 0.8 I find 106 families of 10 or more artefacts. The 50 largest cover 18% of the sample, and the rest is a long tail of near-unique scripts. At 0.6, one shared-scaffolding family covers 52%. This is consistent with Section 4.2: style links artefacts to a shared toolkit or configuration, not to one of ~1,200 agents.

5. Discussion and Limitations

Implications.

1. Any identifier an agent controls offers no accountability against an agent that wants to shed it. 2. Behavioural fingerprints identify *configurations*. Against a swarm of replicas, which is exactly the incident's setting, they cannot assign an action to an instance. Individual accountability must come from infrastructure: per-instance credentials issued and logged by the gateway, out of the agent's reach. 3. Fingerprints are valuable as a *lead* layer on top of attestation. They flag accounts that pursue the same goal, robustly to style evasion. Their errors are predictable (shared base models, sincere allies), and automatic merging at scale does more harm than good. 4. To achieve a goal, agents have to say things that serve it. That is why goals leak, and also why goals alone cannot distinguish a cartel from a consensus. Telling them apart needs evidence about *control*, such as gateway logs, payment trails or spawn lineage, not more text analysis.

Who can use this, and how. *Gateway and API providers* (model labs, cloud platforms, MCP gateways) can issue a credential per agent instance and log it, since they are the only party that can tell replicas apart. *Agent platforms and marketplaces* can bind accounts to operators and use fingerprint similarity to queue suspicious accounts for human review, not for automatic bans. *Standards efforts* for agent identity (OAuth/OpenID extensions, A2A) can prefer infrastructure-issued over self-asserted IDs. *Researchers* can reuse the open testbed to test new detectors or ID schemes against agents that rename and hide.

Limitations.

- **The testbed is simplified.** It is one synthetic forum with short posts (median 75 words, about 1.8 posts per alias), 10 rounds, and random rather than strategic renaming. Alias boundaries are clean.
- **Small samples.** There are 2–5 episodes per condition. The coordination–agreement comparison rests on 1–4 front pairs and 1–2 twin pairs per episode, so its confidence intervals are wide.
- **Operator design.** The operators, missions and agendas were written by me. The large design randomises roles and styles but draws from a fixed pool.
- **Replicas by construction.** Agents of one operator are identical by design. Real agents differ through tools, tasks and long histories, which my memory condition only begins to model.
- **Inconsistent reasoning settings.** Frontier models with mandatory reasoning (Claude Sonnet 5.5, Gemini 3.1 Pro) ran at low reasoning effort; the others ran without reasoning.
- **Schemes not implemented.** The gateway and key schemes are scored analytically from ground truth, not built cryptographically.
- **Untested threats.** I did not test adaptive adversaries that optimise against the detector, or LLM-judge detectors.
- **Real data.** The real-data probe has no ground truth.

Future work. Gateway-issued per-instance credentials with spawn lineage and revocation; adaptive evasion; distinguishing coordination from agreement using control signals (timing, shared infrastructure); validation on AI Village transcripts, which carry agent labels.

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Appendix A. Limitations and Dual-Use / Ethical Considerations

What this does and does not show. In controlled simulations I show four things: renamable identifiers carry no accountability; behavioural fingerprints of LLM agents identify operator configurations rather than individual instances; content signals outlast style evasion; and content signals cannot by themselves separate covert coordination from sincere agreement. I do not show that the numbers transfer to real deployments, longer interactions, tool-using agents or adaptive adversaries. The real-data probe is descriptive only.

Dual use.

1. **Evasion guidance.** The evasion results tell a covert operator which strategies work best: paraphrasing through another model is the strongest. I judge this low-risk, because these strategies are obvious and already in use, and I also show their limit (goals leak). 2. **De-anonymisation and false accusation.** Fingerprinting can de-anonymise legitimate pseudonymous agents or operators. More importantly, my honest-twin result shows it would **falsely accuse sincere allies of coordination**. Fingerprint evidence should therefore trigger human review with due process, never automatic enforcement. Shared base models are a known source of false matches. 3. **Containment.** The covert agendas promote fictional, harmless products. Agents had no tools, no network access and no effects outside the simulation. No cyber-offensive content was generated. Swarm Traces payloads were processed as text only and never executed.

Use of AI tools. Code, analysis and drafting were assisted by an AI coding assistant. Every number in this report is produced by the released code from logged runs, and every reference was checked against its arXiv record or publisher page.

Appendix B. Reproducibility and run inventory

Code, prompts, every post with ground-truth labels, all metrics and the API response cache are in the project repository (`whoiswho/`, `results/`). The total API cost of all 62 swarms, embeddings and audits was about $12.60 via OpenRouter. Settings: temperature 0.9; reasoning disabled, or effort "low" where a provider requires reasoning; embeddings from `openai/text-embedding-3-small`; paraphraser `meta-llama/llama-3.3-70b-instruct`.

Design	Conditions (episodes)	Agents per swarm
Small, fixed roles	no evasion (5), persona (5), paraphrase (5), model mixing (5), no style (3), subtle (3), memory (3), hard (3), frontier (2), frontier + persona (2)	40
Small, randomised roles + honest twins	obvious (5), subtle (5)	40
Large, randomised roles + twins	obvious (5), subtle (5), persona (3), memory (3)	160

Table 3. All 62 swarms: 4,400 agents, 44,000 posts, 24,441 aliases.

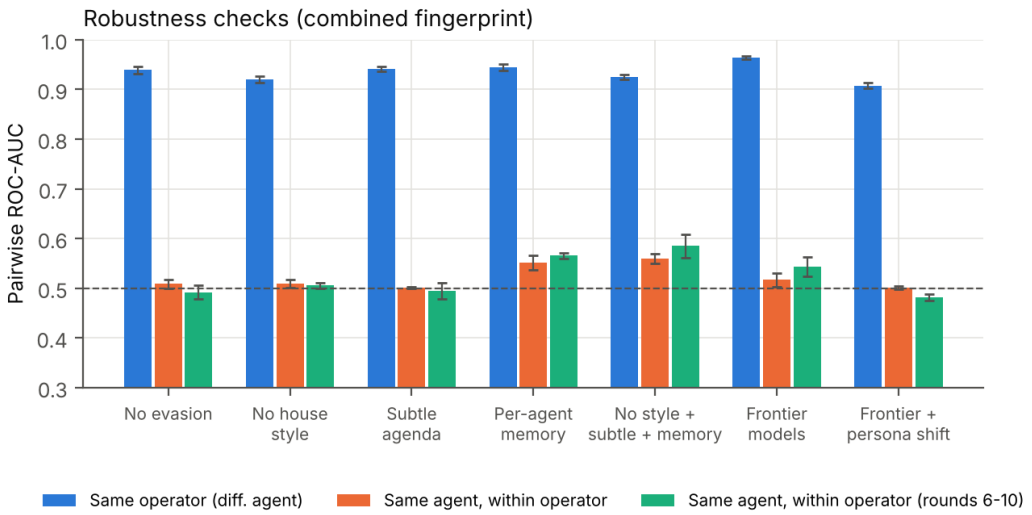


Figure 3. Robustness checks (combined fingerprint): same operator, same agent within an operator, and the latter for aliases in rounds 6–10 only.

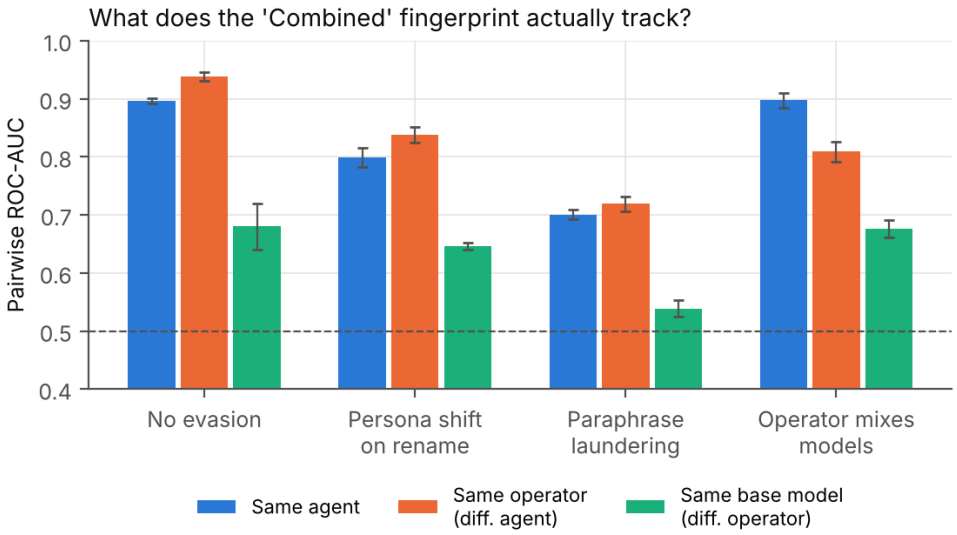


Figure 4. What the combined fingerprint tracks in the small fixed-role design: same agent, same operator (different agents), same base model (different operators).

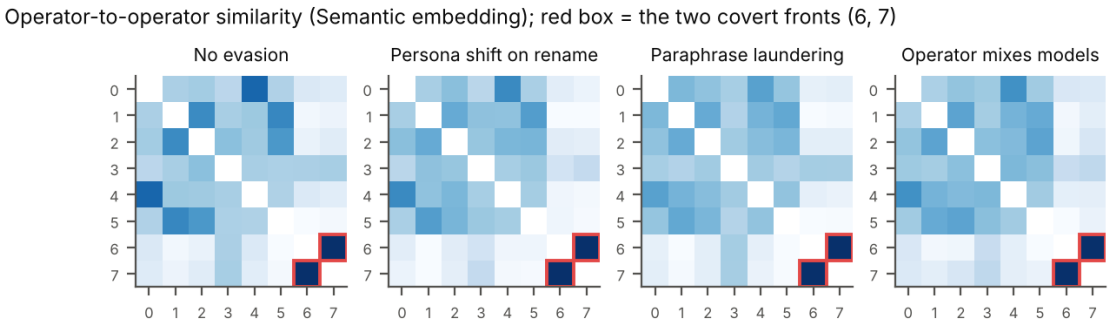


Figure 5. Operator-to-operator semantic similarity in the small fixed-role design (seed 0). The covert fronts (6, 7) are the most similar pair. The fainter dark cells (0–4, 1–5) are operators sharing a base model.